

# Junyi (Conny) Zhou

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## EDUCATION

- **Harvard University, T.H. Chan School of Public Health** Boston, MA  
*S.M. in Health Data Science* August 2025 – Expected December 2026
  - Coursework: Deep Learning, Healthcare Machine Learning
- **Emory University, College of Arts and Sciences** Atlanta, GA  
*B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925* August 2021 – May 2025
  - Coursework: Data Structures, Database Systems, Machine Learning, Natural Language Processing

## TECHNICAL SKILLS

- **Languages:** Python, C++, Java, JavaScript, SQL, R
- **Software:** Flask, React, REST APIs, Auth0, Git, MongoDB, PostgreSQL, MySQL
- **AI/ML:** PyTorch, OpenAI APIs, RAG, scikit-learn, Hugging Face Transformers
- **Cloud & Tools:** AWS Elastic Beanstalk, CloudFront, EC2, S3, Jupyter, Cursor

## SOFTWARE ENGINEERING EXPERIENCE

- **Airway Management Simulation Platform** Atlanta, GA  
*Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital* January 2024 – August 2025
  - Architected a **React** client and **Python/Flask REST API** backed by **MongoDB**, with **Auth0** authentication for a clinical training platform.
  - Deployed the Flask backend through **AWS Elastic Beanstalk** and delivered the frontend through **CloudFront**, configuring SSL and DNS for browser access.
  - Integrated an **OpenAI RAG pipeline** with context management and hallucination controls to generate multi-turn simulation scenarios.
  - Led an Agile team of 6, conducting code reviews and incorporating clinician feedback to iteratively improve the application and NLP pipeline.

## TECHNICAL PROJECTS

- **Transformer Architecture Reimplementation** Atlanta, GA  
*Machine Learning Researcher — Emory Dept. of Computer Science* October 2024 – January 2025
  - Implemented core Transformer algorithms from first principles in **PyTorch**, including multi-head self-attention, positional encoding, layer normalization, and the training pipeline.
  - Tuned and benchmarked the implementation against `nn.Transformer`, validating correctness and achieving a **9.38 BLEU** score on English-to-German translation.
- **Parallel AlphaFold Inference Pipeline** Atlanta, GA  
*Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry* February 2025 – August 2025
  - Automated high-throughput inference on HPC clusters, using parallel scheduling to distribute structural predictions across GPU nodes.
  - Built **Python** scripts and CLI tools to filter and rank outputs using pLDDT and PAE scores, with documentation for non-ML researchers.
- **CellOT Cross-Species Modeling** Boston, MA  
*Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute* February 2026 – Present
  - Implemented a custom **PyTorch CellOT** model and **scanpy** data pipelines for latent-space modeling, automated labeling, and single-cell dataset integration.